FINANACIAL TRANSACTION AND LOAN DASHBOARDS

REPORT DATASET 4

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POWER BI ACTIVITY REPORT

Financial Transactions and Loans Dashboard

1. INTRODUCTION

Microsoft Power BI is a data visualisation and business intelligence tool used to convert raw data into meaningful information. It helps users organise data, perform calculations, create interactive charts, and prepare reports for better analysis.

In this activity, Power BI was used to develop a **Financial Transactions and Loans Dashboard** using the dataset provided by the faculty. The activity involved importing data, cleaning and preparing tables, creating relationships, developing DAX measures, and designing interactive visualisations. The dashboard helps analyse customer details, account information, loan status, and financial transactions in a clear and organised manner.

2. OBJECTIVES OF THE ACTIVITY

The main objectives of this activity were:

- To import and prepare the given dataset using Power BI.
- To identify and correct data quality issues such as duplicates, missing values, and incorrect data types.
- To create relationships between different tables.
- To calculate important financial measures using DAX.
- To develop a dashboard using different charts, tables, and KPI cards.
- To use slicers, drill-down, tooltips, and conditional formatting to make the report interactive.
- To understand how Power BI can help analyse financial information efficiently.

3. DATASET DESCRIPTION

The dataset provided for this activity contained 11 tables related to banking and financial operations. These tables contained information about customers, bank accounts, loans, transactions, branches, addresses, and different categories and statuses.

The main tables used in the activity were:

- **Customers:** Contains customer identification and personal details.
- **Accounts:** Contains account IDs, customer IDs, opening dates, and account balances.

- **Loans:** Contains loan details, principal amounts, interest rates, and dates.
- **Transactions:** Contains transaction IDs, transaction amounts, dates, descriptions, and related account details.
- **Account Types and Account Statuses:** Used to classify accounts according to their type and status.
- **Customer Types:** Used to categorise customers.
- **Loan Statuses:** Used to identify different loan statuses.
- **Transaction Types:** Used to classify financial transactions.
- **Addresses and Branches:** Provide location and branch-related information.

These tables were connected to analyse the information together instead of examining each table separately.

[Insert Screenshot 1: Dataset tables imported into Power BI]

4. DATA IMPORTING AND CLEANING

The first step was to import all 11 tables into Power BI. The Transform Data option was used to open Power Query Editor, where the data was checked and prepared before creating the dashboard.

The following data-cleaning activities were performed:

4.1 Removing Duplicate Records

Duplicate IDs were removed from the Accounts, Addresses, Customers, and Loans tables. This helped maintain the uniqueness of important identification fields and reduced the possibility of incorrect counts.

4.2 Handling Missing Values

Missing and invalid values were handled according to the requirements of each table. In the Addresses table, missing street, city, and country values were replaced with suitable labels such as “Unknown Street,” “Unknown City,” and “Unknown Country.”

In the Customers table, missing first and last names were replaced with “Unknown.” The date of birth column was also corrected, and invalid date values were replaced with null values. These records were retained because the customers could still have active bank accounts.

4.3 Correcting Data Types

The data types of important columns were changed to make them suitable for calculations and analysis. For example, opening dates and transaction dates were

converted into the Date format, while account balances, loan principal amounts, and transaction amounts were converted into Fixed Decimal Number format.

4.4 Cleaning Text Data

The description column in the Transactions table was cleaned using the Trim and Clean options. These steps removed unnecessary spaces and unwanted characters that could affect the quality of the data.

After completing the cleaning process, the prepared data was applied to Power BI for further modelling and report creation.

[Insert Screenshot 2: Data cleaning in Power Query Editor]

5. DATA MODELLING AND RELATIONSHIPS

After cleaning the data, relationships were created between the tables using their common identification columns. A total of 12 relationships were specified in the working notes.

For example, Customers was connected to Accounts through CustomerID, while Account Types was connected to Accounts through AccountTypeID. Similarly, Transaction Types was connected to Transactions through TransactionTypeID.

Most relationships used one-to-many cardinality and single-direction cross-filtering. Some relationships were kept inactive because multiple possible connections existed between tables. The relationship between Accounts and Transactions through AccountDestinationID was inactive, allowing a separate measure to calculate transactions involving destination accounts.

An automatically created relationship between Branches and Addresses was also removed as instructed.

Correct relationships were important because they allowed information from different tables to work together when calculations, charts, and slicers were used.

[Insert Screenshot 3: Model view showing relationships between tables]

6. CREATING DAX MEASURES

Data Analysis Expressions (DAX) was used to calculate important values from the dataset. These measures helped display summary information and supported the dashboard visualisations.

The following measures were created:

Measure	Purpose

Total Customers	Counts the distinct customer IDs.
Total Accounts	Counts the distinct account IDs.
Active Accounts	Counts accounts with Active status.
Total Account Balance	Calculates the sum of account balances.
Total Loans	Counts the distinct loan IDs.
Active Loans	Counts loans with Active status.
Total Loan Principal	Calculates the total loan principal amount.
Average Interest Rate	Calculates the average interest rate of loans.
Total Transactions	Counts the distinct transaction IDs.
Total Transaction Value	Calculates the total value of transactions.

For example, the Total Customers measure used the DISTINCTCOUNT function to count unique customers.

Total Customers = DISTINCTCOUNT(Customers[CustomerID])

The Total Transaction Value measure used the SUM function:

Total Transaction Value = SUM(Transactions[Amount])

The CALCULATE function was used to calculate active accounts and active loans based on their status. The USERELATIONSHIP function was also used to create special measures for destination transaction value and total loan amount.

A temporary table visual was created to check the measures and identify any blank results. After validation, the temporary table was removed.

7. DASHBOARD CREATION AND VISUALISATIONS

A report page was created in Power BI and renamed **Finance Overview**. A text box containing the title “Financial Transactions & Loans Dashboard” was added to the page.

Different visualisations were then created to present the financial information clearly.

7.1 KPI Cards

Four card visuals were created to display Total Customers, Active Accounts, Active Loans, and Total Transaction Value.

These cards provide a quick summary of important information without requiring users to examine individual tables. The formatting was adjusted to display whole numbers clearly and show the transaction value using the appropriate currency format.

[Insert Screenshot 4: Finance Overview dashboard with four KPI cards]

7.2 Accounts by Account Type – Clustered Column Chart

A clustered column chart was used to compare the number of accounts under different account types. Account Type was placed on the X-axis, while Total Accounts was placed on the Y-axis.

This visual helps users compare account categories and identify which types have more or fewer accounts.

7.3 Transactions by Month – Line Chart

A line chart was created to display the number of transactions across months. The month from the transaction date hierarchy was placed on the X-axis, and Total Transactions was placed on the Y-axis.

This chart helps users observe changes in transaction activity over time. The blank month category was removed where necessary, and the axis and line formatting were adjusted for clearer presentation.

7.4 Loans by Loan Status – Donut Chart

A donut chart was used to represent the distribution of loans by status. Loan Status was placed in the legend, and Total Loans was used as the value.

Category and percentage labels were enabled to make it easier to understand how the total loans were distributed across different statuses.

7.5 Transactions by Type – Clustered Bar Chart

A clustered bar chart was created to compare the number of transactions across different transaction types. Transaction Type was placed on the Y-axis, and Total Transactions was placed on the X-axis.

Data labels were enabled to make the values easier to read and compare.

7.6 Transaction Summary – Table Visual

A table visual was created to display transaction types, total transactions, and total transaction value. The headings were formatted in bold to improve readability.

This visual provides a detailed summary of transaction activity alongside the charts used in the dashboard.

[Insert Screenshot 5: Dashboard showing charts and transaction summary]

8. ADDING SLICERS AND INTERACTIVITY

Slicers were added to allow users to view specific parts of the dataset without changing the original data.

Three slicers were created:

- **Year Slicer:** A new Year column was created from TransactionDate using the YEAR function. This allowed the user to filter the dashboard by year.
- **Account Type Slicer:** Allowed users to select a particular account category.
- **Transaction Type Slicer:** Allowed users to select a specific transaction category.

The account type slicer was changed to a dropdown style. The other slicers were also configured to make selections convenient.

When the Business account type was selected, the dashboard visuals could be examined to understand how the selection affected accounts, transaction trends, transaction types, the summary table, and KPI cards.

The Edit Interactions feature was also used to control how one visual affected other visuals. The available options included Filter, Highlight, and None. This helped ensure that visual interactions worked as intended.

[Insert Screenshot 6: Dashboard with Year, Account Type, and Transaction Type slicers]

9. DRILL-DOWN FEATURE

Drill-down was used to examine transaction activity at different levels of time. The Transactions Over Time visual was configured with a date hierarchy containing Year, Quarter, and Month.

At the year level, users could view transaction activity for each year. Drilling down to the quarter level provided a more detailed view of activity within each year. Further drilling down to the month level showed the transaction pattern for individual months.

The drill-down mode also allowed a specific year and quarter to be selected for more detailed analysis. The Drill Up option was used to return to a higher level.

This feature made it easier to move from an overall view of transaction activity to more detailed information when required.

[Insert Screenshot 7: Drill-down view showing year, quarter, and month]

10. REPORT PAGE TOOLTIP

A separate report page named **Transaction Tooltip** was created to provide additional information when users interacted with the Transactions by Type chart.

The page was configured as a tooltip page by enabling the Allow Use as Tooltip setting and changing the canvas type to Tooltip.

Three card visuals were added to display:

1. Transaction Type
2. Total Transactions
3. Total Transaction Value

The tooltip page was then connected to the Transactions by Type visual. This allowed users to see additional transaction details when interacting with the relevant data points, without overcrowding the main dashboard.

11. CONDITIONAL FORMATTING

Conditional formatting was applied to the Transaction Summary table, specifically to the Transaction Value column.

Data bars were added to make differences between transaction values easier to identify visually. Green was used for positive values, while red was used for negative values.

This formatting helped users recognise positive and negative values quickly and made the table easier to interpret.

[Insert Screenshot 8: Transaction Summary with conditional formatting]

12. OBSERVATIONS AND LEARNING OUTCOMES

The completed dashboard brought customer, account, loan, and transaction information together in one report. The KPI cards provided a summary of important measures, while the charts helped compare account types, transaction categories, monthly activity, and loan statuses.

The slicers allowed the information to be examined according to selected years, account types, and transaction types. Drill-down provided more detailed views of transaction activity, while tooltips and conditional formatting improved the presentation of additional information.

Through this activity, I gained practical experience in importing data, cleaning records, correcting data types, creating relationships, and writing DAX measures. I also learned how different visualisations can be selected according to the type of information being analysed.

The activity helped me understand that a dashboard is not only a collection of charts. Data preparation, correct relationships, accurate calculations, and appropriate formatting are all important for producing a useful report.

13. CONCLUSION

This Power BI activity provided practical knowledge of developing an interactive financial dashboard using a multi-table dataset. The activity covered the main stages of data preparation, modelling, calculation, visualisation, and report formatting.

The final dashboard presented information about customers, accounts, loans, and transactions through KPI cards, charts, tables, slicers, drill-down, tooltips, and conditional formatting. Overall, the activity improved my understanding of Power BI and showed how it can be used to organise financial data and present it in a clear and meaningful way.

THANK YOU